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Palaeomagnetism of the Roraima Dolerites

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Summary

Consistent palaeomagnetic data have been obtained from 19 out of 31 sampling sites in the thick Proterozoic dolerite sills which intrude the Roraima sandstones of the Guiana Shield in Venezuela and British Guiana (now Guyana). Fourteen of these sites, most of which are in noritic dolerite, have remanence orientations which fall into two significantly different groups, approximately opposite in declination, but both with positive inclination: group I, $N = 7$, $D = 334$, $I = +27$, $K = 29$, $\alpha 95 = 11$, N pole co-ordinate = 63 N 129 W; group II, $N = 7$, $D = 145$, $I = +39$, $K = 25$, $\alpha 95 = 12$, N pole co-ordinates = 45 S, 13 W. The remaining stable sites, mostly from norite which occurs near the base of some thick sills, have scattered orientations. The unstable sites are characteristically in rocks representing later, iron-enriched stages in the dolerite differentiation sequence.

The published potassium–argon ages on these dolerites, and minerals separated from them, give evidence of differential argon loss, but all with a minimum age of about 1500 m.y. There is no clear evidence in the radiometric data of the two separate periods of dolerite intrusion implied by the palaeomagnetism.

An additional four sites, group III, are from dykes which are known or inferred to belong to a distinctly younger dolerite suite, of uncertain age. The mean orientation of these four sites is: $D = 17$, $I = +21$, $K = 36$, $\alpha 95 = 16$, N pole co-ordinates = 73 N, 11 E.

Ilmenite is the exclusive or predominant oxide accessory seen optically in the magnetically stable Roraima dolerites, whereas magnetite is conspicuous in the unstable sites. Magnetite, however, in very small amount, is the only ferrimagnetic phase indicated by bulk magnetic property measurements (Js/T, J/H) of stable samples. It is presumed to be the carrier of the remanence, and its stability as compared with that (much more abundant) in the unstable sites is tentatively attributed to finer grain size. In the stable, younger, group III dolerites, however, magnetite is the predominant accessory oxide.

Introduction

The Roraima formation (McConnell *et al.* 1964) consists of a tabular succession of sandstones and shales up to 7000 ft thick which once covered over 100 000 km^2 in the centre of the Guiana Shield (see Fig. 1). The formation has been intruded at several horizons by dolerite sills up to 1200 ft thick (Bellizzia 1957; Hawkes 1966a).

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Dolerite samples from sills in British Guiana have been dated radiometrically b McDougall *et al.* (1963) and by Snelling (1963). McDougall *et al.* (1963) determine the K–Ar and Rb–Sr isotopic composition of the whole rock, and separated pyroxene plagioclase and potassium feldspar fractions in samples from four localities, an concluded that some sills are at least 2090 m.y. old. Demonstrable loss of radiogeni argon and strontium in some samples, however, precluded positive determinatio of whether or not other sills were significantly younger (1500–1700 m.y.). Snellin (1963, p. 1079) determined K^{40}–Ar^{40} content of a pyroxene-hornblende concentrat from dolerite and separated biotite and muscovite fractions from the adjacent contac hornfels. The weighted mean of his results gave an age of 1710 m.y. for the dolerite McConnell *et al.* (1964) do not accept the older age reported by McDougall an reaffirm a radiometric age of circa 1700 m.y. for all Roraima dolerites (their 'younge basic intrusive group'). These determinations at least provide a minimum age for th Roraima formation intruded by the dolerites.

The discovery of pollen and spores of probable Tertiary age in hornfelsed sedi ments of the Roraima formation (Stainforth 1966) is evidence, which if accepte at face value, is in direct conflict with the radiometric ages. The metamorphose state of the rocks containing the pollen, however, cause some geologists to suspec contamination. Nevertheless, as Stainforth reports (*op cit.*, p. 294), differences c opinion persist as to the significance of this pollen.

The freshness of the dolerites, coupled with the radiometric age data, encourage palaeomagnetic investigation. With the active collaboration of the Geological Surve Department in British Guiana, and the Minesterio de Minas e Hidrocarburos i Venezuela, a collection was made in January and December, 1965 of 181 oriente cores of dolerite from 35 separate sites (see Fig. 1 and Table 1).

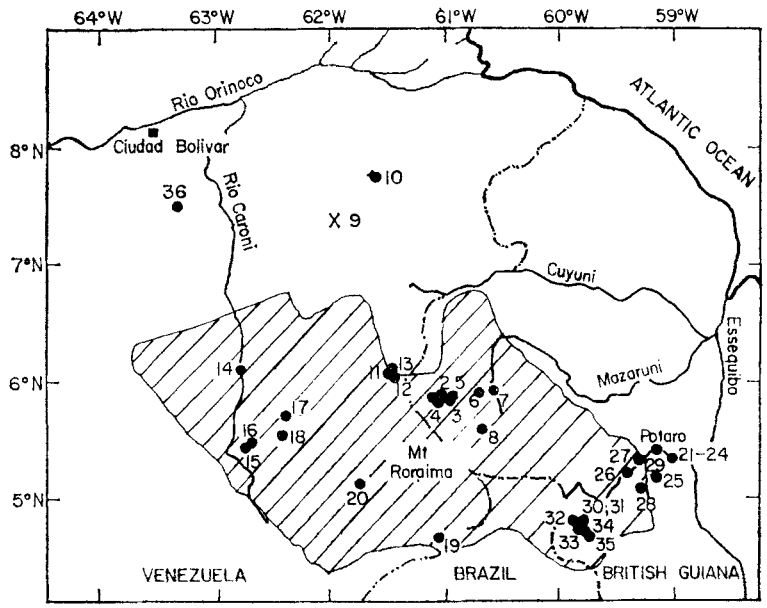


FIG. 1. Map showing distribution of sampling sites (by number) in Guiana Shield. The cross-hatched area indicates the main outcrop area of the Roraima sandstone formation.

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Table 1

Summary of palaeomagnetic data for each site after 300 *to* 500 *Oe a.f. demagnetization. Petrological type classifications are according to Hawkes* (1966*a*); *the bulk susceptibility* (χ) *in e.m.u./cm*^{3} *Oe, No. is the number of sample cores drilled at each site, N is the number of sample cores used in the analysis, and n is the total number of specimens measured, D = declination, I = inclination, K = Fisher's precision parameter, and* $\alpha 95$ = *the cone of* 95 *per cent confidence*

Site number	Site co-ordinates	Site name	Petrological type	χ	Location of body	No.	N	n	D	I	K	α95	R
1	5° 49′ N, 61° 03′ W	Ilubia Falls	Pigeonite dolerite	$6{\cdot}2\times10^{-4}$	Main sill (?)	4	4	11		Random‡			2·93
2	5° 51′ N, 61° 03′ W	Paruima Church	Noritic dolerite*	$2{\cdot}8\times10^{-4}$	Main sill†	4	4	9	330½	14½	10	30	3·71
3	5° 51′ N, 61° 03′ W	Konotipu	Noritic dolerite*	$1{\cdot}6\times10^{-4}$	Main sill	6	6	14	337½	28½	81	8	5·94
4	5° 50′ N, 61° 03′ W	Kawaitipu	Noritic dolerite*	$1{\cdot}5\times10^{-4}$	Main sill	4	4	10	352½	38½	12	28	3·75
5	5° 52′ N, 61° 00′ W	Ituwibisi Falls	Noritic dolerite	$2{\cdot}4\times10^{-4}$	Main sill	8	6	26	329½	33½	90	7	5·94
6	5° 52′ N, 60° 40½′ W	Kamarang 'oxbow'	Noritic dolerite	$8{\cdot}9\times10^{-5}$	Main sill	2	2	4	327	46	259	16	1·99
7	5° 53′ N, 60° 35½′ W	Kamarang airstrip	Noritic dolerite	$3{\cdot}4\times10^{-4}$	Main sill	6	6	12	325½	14	133	6	5·96
8	5° 37½′ N, 60° 43′ W	Chinakuruk	Norite (cumulus)	$9{\cdot}3\times10^{-5}$	High level sill	8	7	23	223	44½	9·5	21	6·37
9	7° 20′ N, 61° 45′ W	El Callao	Pigeonite dolerite	$2{\cdot}2\times10^{-3}$	Minor dyke suite	8	8	17	5	23	35	10	7·80
10	7° 40′ N, 61° 35′ W	Nuria (Quebrada Caballape)	Noritic dolerite*	$1{\cdot}3\times10^{-3}$	Basement: cone sheet	8	8	28	135	57	5·9	25	6·78
11	6° 00′ N, 61° 30′ W	La Escalera, Km 109	Noritic dolerite	$5{\cdot}8\times10^{-4}$	Unconformity sill	8	7	21	144	37	2·9	43	4·93
12	5° 59′ N, 61° 30′ W	Km 112	Noritic dolerite	$2{\cdot}4\times10^{-3}$	Unconformity sill	2	2	6		Widley scattered			
13	6° 00′ N, 61° 30′ W	Km 104	Noritic dolerite	$8{\cdot}4\times10^{-4}$	Unconformity sill	1	1	3		Widely scattered			
14	6° 30′ N, 62° 40′ W	Guance, Rio Caroni	Pigeonite dolerite	$1{\cdot}8\times10^{-3}$	Unconformity sill	4	4	12	294½	74	4·2	51	3·29
15	5° 20′ N, 62° 40′ W	Aprada Ambituir (base)	Noritic dolerite	$3{\cdot}3\times10^{-4}$	Main sill	4	4	18	138½	20½	118	9	3·97
16	5° 20′ N, 62° 39½′ W	Aprada Ambituir (top)	Pigeonite dolerite	$9{\cdot}6\times10^{-4}$	Main sill	4	4	12	152½	51	46	14	3·94
17	5° 44′ N, 62° 16′ W	Kamarata	Hornblende Granophyre	$1{\cdot}4\times10^{-4}$	Main sill	2	2	6		Widley scattered			
18	5° 44′ N, 62° 20′ W	Quebrada Aicha	Noritic dolerite*	$1{\cdot}1\times10^{-3}$	Main sill	6	6	21	334	15	48	10	5·89
19	4° 35′ N, 61° 07′ W	Santa Elena	Pigeonite dolerite	$2{\cdot}4\times10^{-3}$	Basement dyke	4	4	12		Random			1·27
20	5° 03′ N, 61° 48′ W	Uonken	Noritic dolerite	$5{\cdot}5\times10^{-4}$	Main sill	4	4	13	147½	17	166	7	3·99
21	5° 22′ N, 59° 00′ W	Tumatumari N.	Pigeonite dolerite	$1{\cdot}8\times10^{-3}$	Basement dyke	6	6	18		Random			2·43
22	5° 22′ N, 59° 00′ W	Tumatumari S.	Pigeonite dolerite	$1{\cdot}8\times10^{-3}$	Basement dyke	4	4	12		Random			2·33
23	5° 22′ N, 59° 00′ W	Tumatumari W.	Pigeonite dolerite	$1{\cdot}9\times10^{-3}$	Basement dyke	5	5	15		Random			3·33
24	5° 22′ N, 59° 00′ W	Tumatumari W.	Minor dyke suite	$1{\cdot}7\times10^{-3}$		7	4	8	16	13½	10	30	3·70
25	5° 13′ N, 59° 07′ W	Eagle Mountain	Ferrodolerite	$3{\cdot}3\times10^{-3}$	Inclined sheet	5	5	12		Random			1·30
26	5° 17′ N, 59° 23′ W	Waratuk Falls	Metadolerite	$4{\cdot}7\times10^{-5}$	Basement	6	5	10	288	39½	20	18	4·80
27	5° 17½′ N, 59° 22½′ W	Kuruabaru	Norite (cumulus)	$5{\cdot}4\times10^{-5}$	Unconformity sill	8	8	15	327	−18	26	11	7·64
28	5° 08′ N, 59° 08′ W	Northfork Creek	Ferrodolerite	$2{\cdot}0\times10^{-3}$	Inclined sheet	8	8	16		Random			3·33
29	5° 22½′ N, 59° 07½′ W	Potaro Bridge	Minor Dyke Suite	$1{\cdot}8\times10^{-3}$		4	4	8	16½	34½	47	14	3·94
30	4° 57½′ N, 50° 51½′ W	Kopinang Mission (1)	Hornblende Granophyre	$2{\cdot}3\times10^{-3}$	Main sill	6	5	10	244	12	4·0	44	4·00
31	4° 57½′ N, 50° 51½′ W	Kopinang Mission (2)	Ferrodolerite	$1{\cdot}0\times10^{-4}$	Main sill	4	4	9		Random			2·79
32	4° 58′ N, 59° 54½′ W	Kopinang Mountain	Norite (cumulus)	$8{\cdot}1\times10^{-4}$	High level sill	8	8	16	300½	−5½	18	14	7·49
33	4° 56′ N, 59° 50½′ W	Uribaru Tributary	Noritic dolerite*	$4{\cdot}4\times10^{-4}$	Main sill	5	4	8	153½	41	16	24	3·82
34	4° 57′ N, 59° 49′ W	Augustine Falls	Noritic dolerite	$9{\cdot}0\times10^{-5}$	Main sill	4	4	8	142½	50	128	8	3·98
35	4° 55½′ N, 59° 47½′ W	Maikwok	Hornblende Granophyre	$5{\cdot}5\times10^{-4}$	Main sill	4	4	8		Random			2·23
36	7° 28′ N, 63° 22′ W	Cerro Bolivar	Minor dyke suite	$1{\cdot}4\times10^{-3}$		3	3	8	28½	13½	6·0	56	2·67

* Transitional to pigeonite dolerite.

† The palaeomagnetic evidence suggests that what has hitherto been regarded as one 'Main sill', probably consists of more than one intrusion.

‡ Directions are random according to the criterion of Watson (1956).

Petrology

With the exception of the dykes at Cerro Bolivar, El Callao and the Tumatumari vicinity, and the cone-sheet (?) forming the Esplanacia de Nuria (Fig. 1, Sites 36; 9; 21–25, 28 and 29; 10), the remaining dolerites form sills intruding the Roraima formation. The sills occur at the base, and at two general levels within the formation.

Hawkes (1966a, b), in a detailed study of these dolerites distinguishes a Roraima Intrusive Suite, and a younger Minor Dyke Suite. He presents evidence indicating that within the Roraima Intrusive Suite, some basement dykes, the unconformity sill, and higher-level sills in the Roraima Formation are all consanguineous and probably continuous (Hawkes 1966a, p. 324).

The petrological variety of the samples utilized in this study (see Hawkes (1966a)) range from medium-grained norites with cumulus orthopyroxene, through two-pyroxene (orthopyroxene+augite, and pigeonite+augite) dolerites to iron rich varieties containing hornblende, fayalite and interstitial micropegmatite. The suite is typically tholeitic, and the variety is such as is commonly encountered as a result of differentiation in thick dolerite hills.

Some fine grained dolerites are definitely from the Minor Dyke Suite (samples at Site 24 were collected from the younger dykes which cut the main Tumatumari intrusion); others may well be chilled borders from the main Roraima Suite Intrusives.

D. D. Hawkes (personal communication) has kindly examined thin sections representing most of the sites sampled in this study, and classified them according to the various stages in the differentiation sequence they are inferred to represent (see Hawkes (1966a), Table 1, p. 325). The stages he distinguishes range from early differentiates (norites and noritic dolerites), middle (pigeonite dolerite) to late (ferrodolerites). Hornblende granophyres represent heteromorphs of some pigeonite dolerites and ferrodolerites. The sites are classified petrologically in these terms, in Table 1.

The progressive iron enrichment evidenced in this differentiation sequence is compatible with crystallization of basalt magma under a low constant or decreasing fo_2 (Osborn 1959, 1962). That the predominant primary accessory oxide in these rocks is pure ilmenite, as seen in polished section and confirmed by X-ray diffraction studies, is compatible with the high FeO/Fe_2O_3 ratios found (Hawkes *op. cit.*), and called for by this crystallization model.

Sampling and measurement

From two to eight orientated cores, averaging four inches in length, were obtained at each site by means of a portable gasoline-powered diamond drill (Doell & Cox 1967). The cores were oriented *in situ* with the aid of an astro-compass, (La Rochelle 1961). The three oriented hand samples from Cerro Bolivar were drilled in a press and reoriented in a sand-box.

The remanence measurements were made on a spinner magnetometer (Phillips & Kuckes 1967); the alternating-field demagnetization apparatus is described by McElhinny (1963). The Curie Points and saturation magnetization measurements were made on a balance constructed by Collinson (1967). The maximum error, in orientation and measurement, is estimated to be less than 5°.

Results

The palaeomagnetic results are summarized in Tables 1 and 2. Preliminary alternating field (a.f.) demagnetization of samples from several sites, in small increments up to 1000 Oe, showed 300 Oe to be the optimum field for routine demagnetization.

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Table 2

Site-group averages, and equivalent pole positions; pole positions—II are calculated using site co-ordinates after South America is juxtaposed with Africa according to Bullard et al. (1965)

Group	Sites	*D*	*I*	*K*	α95	Pole position Lat.	Long.	*K*	α95	Pole position II Lat.	Long
I	2, 3, 4, 5, 6, 7 and 18	333½	27½	29	11	63 N	129 W	47	9	22 N	94 W
II	10, 11, 15, 16, 20, 33 and 34	145	39½	25	12	45 S	13 W	40	10	14 S	45 E
III	9, 24, 29 and	16½	21	36	16	73 N	11 E	53	13	64 N	83 W
Miscellaneous	8, 14, 26, 27, 30 and 32										

All samples were demagnetised at 150 and 300 Oe. Invariably the grouping at 300 Oe was equal to or better than that at 150 Oe. At some sites, the grouping was further improved by demagnetizing odd samples at 500 Oe. However the unstable sites in Table 1 were not significantly improved by demagnetization in fields as high as 1000 Oe.

Of the 36 sites sampled, the results from 12 (33 per cent) are so widely scattered and inconsistent (despite intensive A.F. demagnetization) as to be useless. The magnetic and petrological characteristics of stable vs unstable samples are compared in a later section. The results for the remaining 24 sites are illustrated in Fig. 2. Two distinct groups are evident, with declinations NW and SE and normal inclination. Four sites have NNE declination, and several are scattered.

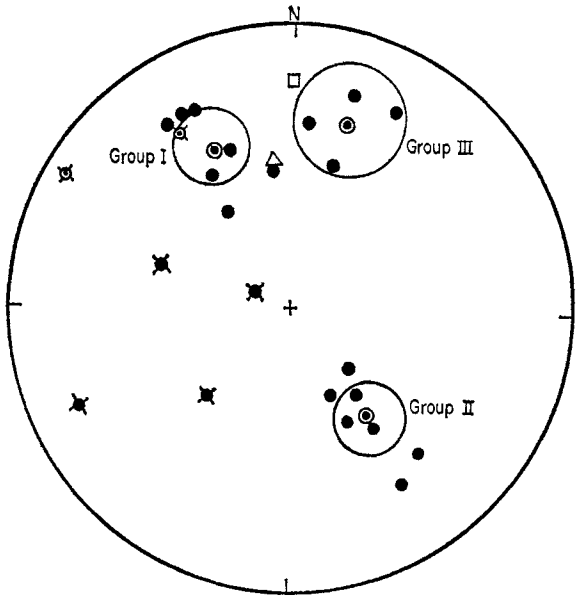


FIG. 2. Stereographic projection of all site averages for which significant consistency was obtained. Solid circles: north-seeking pole on lower hemisphere, open circles on upper hemisphere. The site group averages (I, II, and III) are shown by double circles, together with their cone of 95 per cent confidence. The circle with crosses signify the Miscellaneous sites, none of which are included in the Site Group averages. Open square is the dipole field, the triangle is the mean geomagnetic field.

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Discussion

To facilitate discussion, the site-mean directions plotted in Fig. 2 are grouped as in Table 2.

Groups I and II. The sites in these groups (see Fig. 1) are of the Roraima Suite proper, and excepting the cone sheet at site 10, represent sills completely within the Roraima formation or at the basal unconformity. The bulk belong to the category 'Main Sill', intruded roughly midway in the Roraima sequence.

Petrologically, they range from noritic dolerites, to pigeonite dolerites, the more extreme differentiates being excluded.

Although almost 180° opposite in declination, the inclination of both groups is positive. Even were one to be reversed, the mean vectors would still be over 60° apart, which is a significant difference (Watson & Irving 1957).

Group III. The Minor Dyke Suite (Hawkes 1966a) is described as being constituted of a swarm of narrow (less than 500 ft thick) dykes of prevailing northeast strike which are widespread throughout the Guiana Shield. Hawkes (*op. cit.*, p. 334) suggests a Mesozoic or early Tertiary age for the suite. However, an age of 450 million years was obtained by K–Ar method on both plagioclase and pyroxene separated from a dolerite dyke cutting the South Savana Granite at co-ordinates: 2° 47′ N, 59° 32′ W. (C. N. Barron, personal communication). This dyke has an ENE strike and may be correlative with the Minor Dyke Suite.

Of the sites in Group III, No. 24 is unequivocally a member of the Minor Dyke Suite. It consists of several narrow dykes, which strike NE and cut the older main Tumatumari dyke (Hawkes 1966b). Site 29 cuts basement gneisses, but also strikes NE and was correlated with the Minor Dyke Suite by the Geological Survey even before the palaeomagnetic data were obtained.

Site 9 is a dyke 300 ft wide exposed underground in the Laguna mine, at El Callao. There is no fresh outcrop, but the dyke causes a distinct aeromagnetic anomaly which extends for a considerable distance to the northeast. (J. O. Kalliokoski, personal communication). On the basis of this NE strike, narrow width, and palaeomagnetics (Table 1) this dyke is assigned to Group III and correlated with the Minor Dyke Suite.

Site 36 is from a prominent dyke which cuts the basement at Cerro Bolivar. It extends for over 100 km in a north-easterly direction (Kalliokoski 1965, Plate 1). On the basis of its strike, and palaeomagnetism (Table 1), it is included in Group III.

Miscellaneous

The rock sampled at Site 26 is a metamorphosed basic to intermediate volcanic or shallow intrusive, composed of albite, chlorite, epidote, with traces of amphibole, and with relic igneous textures. Some completely altered titaniferous magnetite grains can be recognized. It forms a ledge in the basement responsible for the Waratuk falls or rapids on the Potaro River. The rock is magnetically stable, and gives a consistent vector of quite distinct orientation. As this is obviously not a fresh dolerite, however, its palaeomagnetism need not necessarily correlate with those of the Roraima intrusives. Its temporal significance with respect to the Roraima intrusives is unknown.

The three anomalous sites 8, 27 and 32 represent the only norites sampled, located towards the base of thick sills.

The sill at site 27, Kuruabaru or Waracobra is exposed in a stream tributary to the Potaro River. Its petrology and structural relations have been described by Rust (1963). It is generally sheet-like, floored by basement gneisses, but a short distance west of the sampling site, it abuts laterally with vertical contact against Roraima sandstones.

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The stable remanence at this site has a northwest declination with negative inclination, almost precisely opposite to the mean of Group II, to which it may be related either by field or self-reversal.

Sites 8 and 38 are the only two in the High Level Sill collected. The sites were located at the base of high sheer cliffs, toward the bottom of the sill, which is consistent with their noritic mineralogy. Their average vectors are quite different from one another, both being distinct from the main Groups I and II. The significance of these anomalous directions is unknown.

The orientation of the two remaining anomalous sites, 14 and 30, is likewise not understood, but with very low precision parameters ($k = 4{\cdot}2$ and $4{\cdot}0$ respectively) these results are of dubious significance.

Mineralogy, magnetic properties and stability characteristics

Fourteen* of the 36 sites sampled gave highly inconsistent results, 15 were very consistent, and 7 moderately consistent. Analysis of the petrological and mineralogical data, and investigation of bulk magnetic properties was directed specifically to the search for possible differences between the dolerites which gave consistent versus inconsistent palaeomagnetic data.

Petrologically, the stable Group I and II sites correlate with the earlier products of the differentiation sequence (norites, through noritic dolerites transitional to pigeonite dolerite). These, in polished section, and by X-ray, have almost pure ilmenite as the exclusive or predominant oxide accessory. The unstable sites, on the other hand, correlate with the later stages of the differentiation sequence (pigeonite dolerite, ferrodolerite etc.) which, as Hawkes (1966a, b) has pointed out, is characterized by increasing iron enrichment. In reflected light, magnetite is much more common in these rocks than in the earlier differentiates.

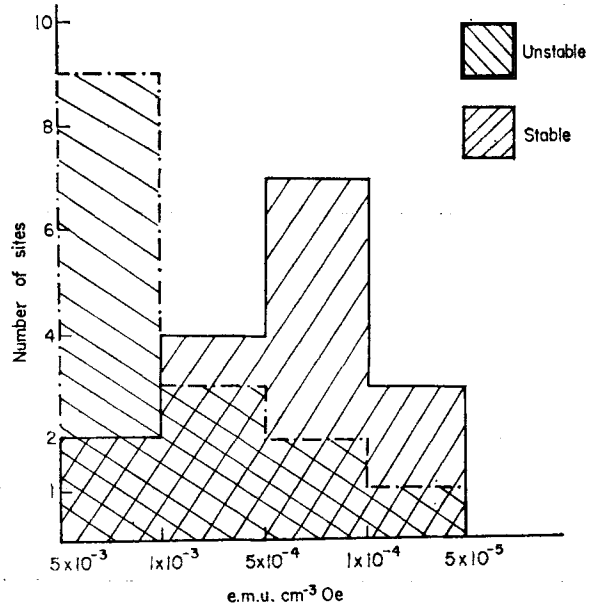


FIG. 3. Histogram to compare the stable and unstable sites with respect to their average bulk susceptibility. Note that the unstable sites characteristically have relatively high values.

* Sites 14 ($k = 4{\cdot}2$) and 30 ($k = 4{\cdot}0$) are included in this total.

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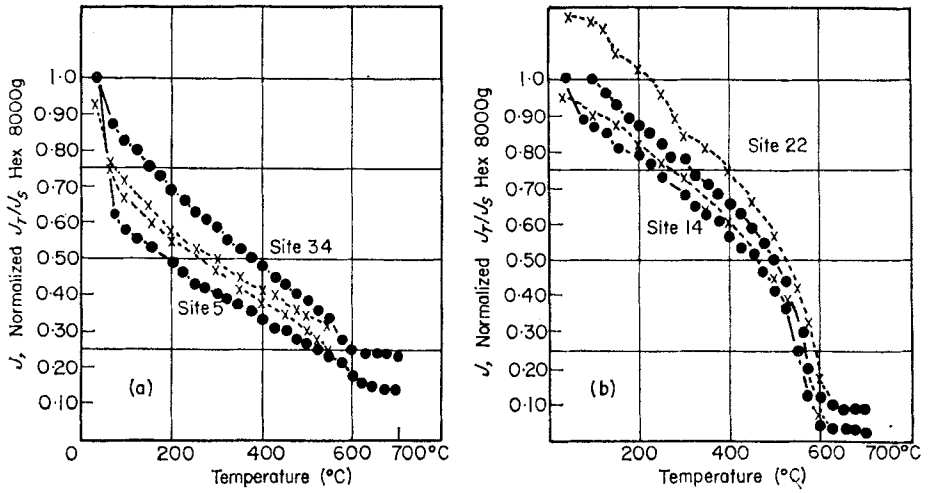


FIG. 4. Representative Js/T curves (Hex = 8000 Oe) for samples from (a) stable, and (b) unstable sites; solid line: heating, broken line: cooling. Note the distinct evidence of magnetite in the unstable samples.

The average bulk susceptibility of all sites is shown in Table I, Fig. 3 is a simple histogram showing the distribution of stable vs unstable sites as a function of bulk susceptibility. The four sites composing group III have high susceptibility (10^{-3} e.m.u./cm^{-3} Oe). These are of uncertain age, probably much younger than the Roraima Intrusive Suite, and are omitted from Fig. 3. Of the remainder, it is very clear that the unstable sites characteristically have much higher bulk susceptibility than the stable sites, in keeping with their greater magnetite content.

Curie point, and saturation magnetization measurements were made on samples from 12 sites, and representative Js/T and J/H curves for stable and unstable samples are illustrated in Figs 4 and 5, and comparative a.f. demagnetization curves in

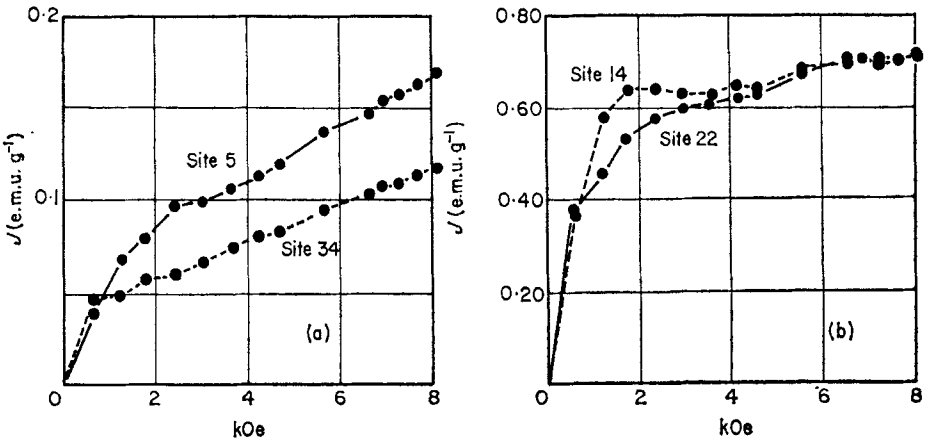


FIG. 5. Saturation magnetization curves for (a) stable and (b) unstable sites using the same samples as in Fig. 4. The linear increase in the intensity of the stable samples above 4000 Oe is attributed to the paramagnetism of the pyroxene. (Note the contrasting intensity scales in (a) and (b)).

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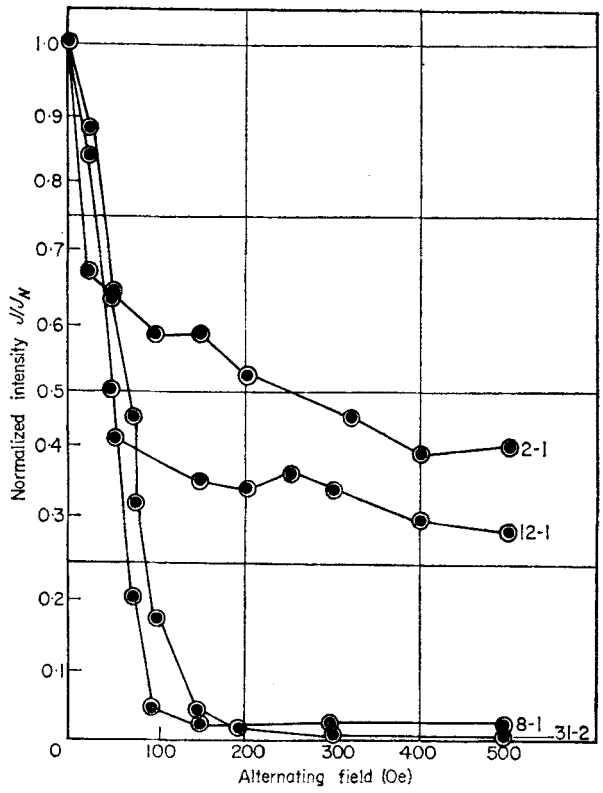


FIG. 6. Representative A.F. demagnetization curves comparing stable (2–1, Site 5 and 12–1, Site 20) and unstable (8–1, Site 14, and 31–2, Site 22) samples.

Fig. 6. The contrast is striking, with magnetite being the dominant magnetic phase in rocks from the unstable sites.

Interpretation of the data from the stable sites is more difficult. The Js/T curves show no conspicuous Curie Points, but a slight drop in intensity is usually evident between 550° and 600 °C suggesting magnetite. Likewise the saturation curves have a slight inflection between 1000 and 3000 Oe suggesting saturation of one component, possibly magnetite. Above about 4000 Oe, the moment of the stable samples increases linearly up to the maximum field available (8000 Oe). Over this range the calculated susceptibility of these rocks varies from 1 to 4×10^{-5} e.m.u./g. Oe. The paramagnetic susceptibility of pyroxenes (Nagata 1961, p. 77) is roughly equal to $x.10^{-4}$ e.m.u./g, where x is the mol fraction of $FeSiO_3$; these dolerites contain of the order of 50 per cent by weight augite of which about 30 mol per cent is $FeSiO_3$. It thus seems most probable that the paramagnetic susceptibility of the pyroxene (perhaps together with ilmenite for which there are no data available) is responsible for the observed increase in moment of the rock in high fields. This paramagnetism would also account for the residual magnetism above 700 °C in the Js/T curves (see Fig. 4(a)). This is most marked in the curves of the stable samples, for which a greater quantity of rock powder was used.

A plot of reciprocal susceptibility ($1/\chi$) vs temperature for the stable samples from site 34 is linear up to about 300 °C. Projected to lower temperatures it intersects the T axis below 0 °K rather than precisely at 0 °K as required by the Curie Law for paramagnetics, which suggests the presence of an antiferromagnetic phase (Nagata 1961, p. 21). This could be the effect of the ilmenite, superimposed on that of the pyroxene, but no confirmatory low-temperature measurements were made.

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Magnetite is the only ferrimagnetic phase in the stable samples for which there is evidence in these data. In what way it differs from the more abundant magnetite in the unstable samples is not clear. That it was not identified at all in some polished sections of stable samples, suggests that it is very fine grained, and perhaps present as the opaque dust particles in the silicates. Evans & McElhinny (1966, p. 6061) for example, found a very stable remanence in a gabbro to be due to small elongate magnetite particles 'exsolved on the cleavage planes of the clinopyroxene'. Young & Hargraves (1967) found the most stable remanence in a diabase sample to reside in the plagioclase fraction, rather than in the pyroxene or oxide. In the Roraima dolerites this 'stable' magnetite could be present in all samples, but is obscured by the preponderant, less stable, coarse-grained accessory magnetite in the unstable samples.

The magnetite-bearing 'high' susceptibility dolerites of Group III, give consistent palaeomagnetic data. It is not known whether there is some intrinsic difference in the magnetite of these rocks (they may well have cooled more rapidly), or whether the very much greater age of the Roraima Dolerites have resulted in much greater spontaneous decay of original remanence and hence greater inconsistency.

Discussion of palaeomagnetic results

Even if one were reversed, the mean vectors of Group I and Group II (Fig. 2) are significantly different (Watson & Irving (1957). No simple field reversal or self reversal mechanism can account for the contrast in orientation. Representative samples from both groups are indistinguishable petrologically and in terms of their bulk magnetic properties.

The rocks of the two groups acquired their magnetism at different times, and if, as seems most probable, the magnetism is primary TRM, then the results imply that there are dolerites of two distinct ages in the Roraima Intrusive Suite.

Group III dolerites are distinct palaeomagnetically, and in terms of bulk magnetic properties; their correlation with the Minor Dyke Suite seems reasonable.

Radiometric age

The published data on radiometric ages of the Roraima dolerites were reviewed briefly in the Introduction.

McDougall *et al.* (1963) clearly consider the possibility, in terms of their isotopic data, that there may be two ages of dolerite in the Roraima province, some as old as 2090 m.y. A later thermal-event around 1700 m.y., which caused variable loss of radiogenic Ar and Sr, may have been associated with a second period of dolerite emplacement.

In the course of the first collecting trip of this palaeomagnetic study, only one site (No. 7, Kamarang Airstrip) was in common with the radiometrically dated samples. When palaeomagnetic evidence of two distinct groups was forthcoming the writer successfully sought the collaboration of Dr Ian McDougall in continuing his age work on the palaeomagnetic samples. Concurrently, a second collecting trip was undertaken specifically to sample for palaeomagnetic study localities from which samples had already been dated.

The new data provided by McDougall (personal communication) is briefly summarized in Table 3 (for details, see companion paper by I. McDougall) together with all previous radiometric data. The palaeomagnetic classification (Group I, II, miscellaneous or unstable) of the 'dated' sites is also indicated where available. It is clear that there is no evidence of a significant difference in K–Ar age of the two magnetic groups.

In view of the differential loss of radiogenic argon which has occurred (McDougall *et al.* 1963), it remains possible that two ages of dolerite are present, the older being

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Table 3

Summary of K–Ar age determinations on Roraima dolerites

Reference	Reference No.	Locality name	Mineral dated	Age (m.y.)	Palaeomagnetic Site	Palaeomagnetic Group
(1)	G.A.687	Kamarang Airstrip	i. Plagioclase	1840	7	I
			ii. Pyroxene	1720	7	
(1)	G.A.688	Kamarang Airstrip	i. Plagioclase	1760	7	
			ii. Pyroxene	1690	7	
(1)	G.A.689	Kopinang Sill,	i. Plagioclase	2085		
(4)		at Muribang River	ii. Pyroxene	1775		
(1)	G.A.690	North East Ridge	i. Plagioclase	1600		
			ii. Pyroxene	1670		
(1)	G.A.691	Tumatumari	i. Plagioclase	1660	31	Unstable
			ii. Pyroxene	1540	31	
(1)	G.A.692	Eagle Mountain	i. Plagioclase	1610	32	Unstable
			ii. Pyroxene	1670	32	
(2)	O.G.S.63.7	Contact Hornfels	i. Muscovite	1735		
(2)	O.G.S.63.8	Kopinang Sill	ii. Biotite	1700		
(2)	O.G.S.63.9	Velgraad	Dolerite	1665	40	II
(3)	O.G.S.63.80	Mt Roraima	Dolerite	1570		
(4)	G.A.791	Siparuni Head	i. Plagioclase	1596	32	Miscellaneous
		(Kopinang Mt)	ii. Pyroxene	1660	32	
(4)	G.A.2058	Ituwibisi Falls	i. Plagioclase	1494	5	I
			ii. Pyroxene	1598	5	
(4)	G.A.2059	Aprada Ambituir	i. Plagioclase	1497	15	II
		Venezuela	ii. Pyroxene	2073	15	
(4)	G.A.2060	Quebrada Aicha	i. Plagioclase	1576	18	I
		Venezuela	ii. Pyroxene	1736	18	
(4)	G.A.2061	Uonken, Ven.	i. Plagioclase	1492	20	II
			ii. Pyroxene	1584	20	

(1) McDougall *et al.* (1963).

(2) Snelling (1963).

(3) C. N. Barron, personal communication.

(4) McDougall *et al.* (1963).

rejuvenated at the time of emplacement of the younger intrusives, or both at some completetly later time.

The two magnetic groups are significantly different in orientation, and presumably different in age. Both group magnetizations are at least 1500 m.y. old, but which of the two is the older, and by how much, cannot be established from these data.

These results imply the survival of primary remanent magnetization despite metamorphism (of which there is no petrographic evidence) sufficient to cause leakage of radiogenic Ar and Sr. Jones & McElhinny (1966) obtained comparable evidence from Proterozoic dolerites in Southern Africa: consistent palaeomagnetic orientations from dykes which gave a considerable spectrum in K–Ar age (*op. cit.*, p. 551). These data illustrate the uncertainty in correlations between Pre-Cambrian palaeomagnetic results and companion radiometric age measurements.

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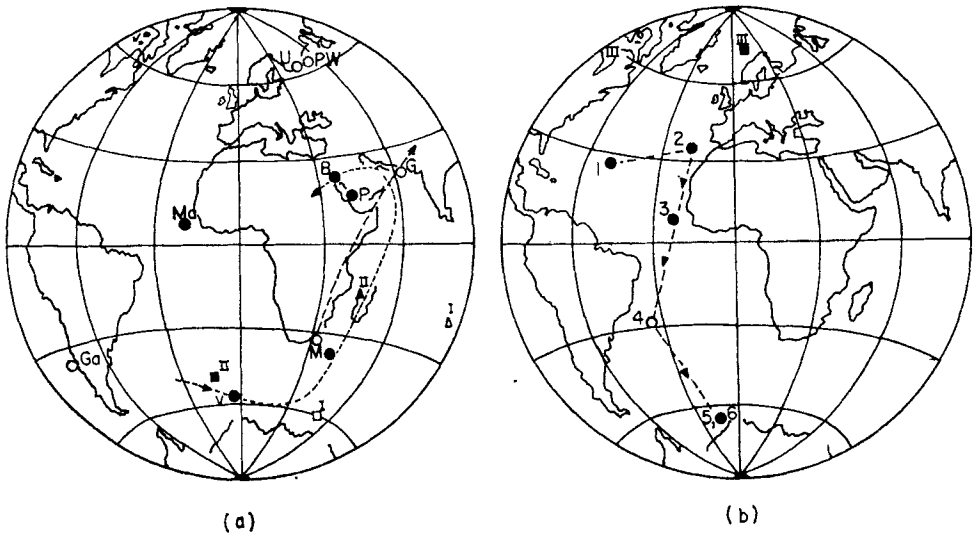


FIG. 7(a). Comparison of the Roraima dolerite poles (squares, Groups I and II) with those obtained from Proterozoic African rocks. The Roraima poles indicated by triangles are those calculated from site co-ordinates when South America is juxtaposed with Africa according to Bullard *et al.* (1965). The dotted and broken lines are segments of the two possible African Precambrian polar wander curves reported by McElhinny *et al.* (1968). Mo = Modipe gabbro, Evans & McElhinny (1966), G = Great Dyke, McElhinny & Gough (1963), B = Bushveld, Gough & Van Niekerk (1959), Ma = Mashonaland, U = Umkondo, McElhinny & Opdyke (1964), P.W. = Post Waterberg, Jones & McElhinny (1966), P = Pilansberg, Gough (1956), Ga = Gaberones, Evans (1967), V = Ventersdorp, Jones *et al.* (1967). Solid figures = N. Poles, open figures = S. Poles.

FIG. 7(b). Comparison of Roraima dolerite Group III pole with the Lower Palaeozoic polar wander curves for South America reported by Creer (1965). 1, Cambrian; 2, Silurian; 3, Devonian, 4; Pennsylvanian; 5 and 6, Permian.

Comparison of results

There are no other Proterozoic palaeomagnetic data from South America with which to make comparison. The present co-ordinates of the virtual poles calculated for Groups I-III are indicated in Table 2. If South America is juxtaposed with Africa according to the fit of Bullard *et al.* (1965), the new poles equivalent to the palaeomagnetic data obtained in this study are also listed in Table 2.

In Fig. 7(a), both poles-sets for Groups I and II are compared with the Pre-Cambrian palaeomagnetic data from Africa. Shown too are segments of the two possible African polar-wander curves (McElhinny *et al.* 1968) with which the Roraima data best compare. Prior to juxtaposition, the Roraima poles are close to the Ventersorp pole (*c.* 2100 m.y.). After justaposition they fall close to both polar wander curves: (a) between Mo (2700 m.y.) and G (2530 m.y.) or (b) between V (2100 m.y.) and B (1950 m.y.). Correlation of the Roraima poles with the latter would at least be in keeping with the conclusion of McDougall *et al.* (1963) that some of the Roraima dolerites are 2090 m.y. old. However it is clear that this correlation is as good before, as after juxtaposition of the two continents.

In Fig. 7(b), the pole for Group III is compared with the Lower Palaeozoic polar wander curve for South America reported by Creer (1965). If Creer's Cambrian pole is correct (No. 1), then it seems most unlikely that the Group III Roraima dolerites can be Ordovician in age. The relative proximity of the Group III pole to the present

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pole suggests that these dolerites may indeed be very much younger: Cretaceous or Tertiary, as favoured by Hawkes (1966a, p. 334). Further consideration is pointless, however, pending direct radiometric determination of the age of the Group III dolerites.

Conclusions

Stable, consistent palaeomagnetic data have been obtained from 19 out of 31 sites in the Roraima Intrusive Suite (excluding four Group III sites, correlated with the Minor Dyke Suite of younger age, and the metadolerite at site 26). Fourteen of these sites fall into two groups of seven sites each, of distinctly and significantly different orientation. Representative samples from both groups are indistinguishable petrologically and in terms of bulk magnetic properties.

This difference in magnetic polarity is considered to be primary, and to date from the time of emplacement and crystallization of sills of two different ages. It has survived a period of heating, possibly associated with the time of emplacement of the younger sills (which ever they may be), which caused substantial leakage of radiogenic argon and strontium from minerals in the rocks, to such an extent that they can no longer be distinguished by radiometric means.

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Palaeomagnetic data obtained from three oriented hand-samples provided by R. L. Pollard, Chief Engineer, Orinoco Mining Company, from a dolerite dyke cutting the basement at Cerro Bolivar have been included in this analysis (site 36).

Dr D. D. Hawkes of the University College of Sierra Leone kindly examined thin-sections of most of the dolerites in order to integrate them with his comprehensive petrological study of Guiana dolerites (Hawkes 1966).

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